SAKTHIVEL S M

AI ENGINEER | MACHINE LEARNING | GENERATIVE AI | LLM & RAG SYSTEMS

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI Engineer with experience building end-to-end solutions across Machine Learning, Deep Learning, Generative AI, LLMs, and Retrieval-Augmented Generation (RAG). Currently developing enterprise AI applications and document intelligence solutions at Ashirvad by Aliaxis. Skilled in Python, model development, retrieval optimization, API development, and AI application testing. Gold Rank holder, Top 5 in AI & Data Science, and national-level hackathon winner with ■4 lakh+ in prizes. Also experienced in technical training, curriculum preparation, and mentoring students through practical AI projects.

TECHNICAL SKILLS

Programming	Python, SQL
Machine Learning	Regression, Classification, Feature Engineering, Data Preprocessing, Model Evaluation, Time-Series Analysis
Deep Learning & Vision	Neural Networks, CNNs, YOLOv8, Image Classification, Anomaly Detection
Generative AI & LLMs	LLMs, RAG Pipelines, Prompt Engineering, Embeddings, Vector Search, Reranking, LLM Evaluation, Hallucination Mitigation
Frameworks & Libraries	FastAPI, React, Next.js, Pandas, NumPy, Scikit-learn
Databases & Retrieval	PostgreSQL, MySQL, Firebase, FAISS, BM25
Tools & Platforms	Git, GitHub, Jupyter, Ollama, REST APIs
Training & Mentorship	AI/ML Instruction, Generative AI Workshops, Curriculum Development, Hands-on Labs, Project-Based Learning, Student Mentoring, Technical Presentations, Coding Assessments

PROFESSIONAL EXPERIENCE

AI Engineer | Ashirvad by Aliaxis, Bengaluru

Jul 2026 – Present

- Developing secure, cost-effective Generative AI solutions for enterprise workflows and internal business processes.
- Building LLM-powered AI assistants and document intelligence systems for company policies and internal knowledge bases.
- Designing RAG pipelines covering document ingestion, text chunking, embedding generation, semantic search, and question answering.
- Combining BM25 keyword retrieval, FAISS vector search, and reranking to improve relevance and retrieval quality.
- Developing backend APIs with FastAPI and user interfaces with Next.js, React, and TypeScript.
- Integrating LLMs to generate context-aware answers grounded in retrieved enterprise documents.
- Implementing authentication, role-based access control (RBAC), session management, automated testing, and reliability checks.
- Evaluating retrieval quality, response latency, and system performance through benchmark tests and iterative optimization.

Technical Trainer — AI & Generative AI | Marcello Tech (Remote)

Jun 2026 – Present

- Delivering instructor-led training in AI, Machine Learning, Python, SQL, and Generative AI to undergraduate students and fresh graduates.
- Preparing structured course plans, presentations, Jupyter notebooks, coding exercises, assignments, and practical project materials.
- Teaching data preprocessing, feature engineering, supervised learning, model evaluation, and deployment fundamentals.
- Explaining LLM concepts including prompt engineering, RAG, AI agents, embeddings, vector databases, and responsible AI practices.
- Mentoring learners through project planning, implementation, debugging, documentation, and final presentations.
- Conducting coding assessments, reviewing project work, and helping learners prepare for technical interviews.

SELECTED PROJECT

SolarMind AI — Intelligent Solar Analytics Platform

Aiseri Lab | Jan 2026 – Jun 2026

- Developed a time-series machine learning pipeline to identify anomalies in solar photovoltaic (PV) system data.
- Used SQL and Pandas for data preparation and feature engineering across structured datasets.
- Integrated an Ollama-based LLM workflow to generate automated reports and actionable data-driven insights.
- Built an end-to-end flow connecting data ingestion, model inference, and user-facing analytics.

GitHub Project

PROFESSIONAL EXPERIENCE — CONTINUED

AI & Machine Learning Trainee | Besant Technologies, Bengaluru
Mar 2026 – Jun 2026

- Completed practical training in Python, SQL, statistics, data preparation, and the machine learning lifecycle.
- Built regression and classification models using linear/logistic regression, decision trees, and ensemble methods.
- Evaluated model performance using accuracy, precision, recall, and F1-score, and compared algorithm results.
- Practiced missing-value handling, categorical encoding, scaling, outlier treatment, and version control with Git/GitHub.
- Studied neural networks and CNN fundamentals through hands-on exercises and collaborative mini-projects.

AI Research Intern — Machine Learning & Deep Learning | IIT Ropar, Annam.ai
Jan 2025 – Dec 2025
Project: Agri-Waste2Fuel | ■3,00,000 prize + ■75,000 funded internship

- Developed Agri-Waste2Fuel, a multimodal classification pipeline combining YOLOv8 and structured inputs; reported 92.4% accuracy and an F1-score of 0.91.
- Created a recommendation engine for biogas, compost, and biofuel pathways using priority scoring and rule-based optimization.
- Developed carbon-emission regression models and evaluated predictions using MAE and RMSE.
- Contributed to the full-stack application using FastAPI, React, PostgreSQL, and Firebase.
- Supported the end-to-end workflow from data preparation and model development to deployment and automated reporting.

Live Project Demo

Student Project Mentor — AI & Machine Learning | Achariya College of Engineering & Technology, Puducherry
2025 – Present

- Mentoring undergraduate teams in AI, Machine Learning, Python, SQL, and Data Science projects for academic and hackathon competitions.
- Guiding teams through problem definition, dataset preparation, model selection, evaluation, deployment, and technical documentation.
- Supporting computer vision, predictive analytics, recommendation system, and Generative AI project development.
- Helping students with architecture choices, debugging, Git/GitHub workflows, presentation preparation, and research documentation.

EDUCATION

B.Tech — Artificial Intelligence & Data Science

Achariya College of Engineering Technology, Puducherry | 2022 – 2026

CGPA: 8.49/10.0 • **Academic standing:** Top 5 in Department

ACHIEVEMENTS & AWARDS

- **Gold Rank:** Top 5 in AI & Data Science, Pondicherry University.
- **IIT Ropar Innovation Challenge:** ■3,00,000 prize and ■75,000 funded research internship.
- **FuturePreneur Grand Finale 2025:** 3rd Place — ■10,000 cash prize.
- **Smart City Hackathon:** Runner-Up — ■10,000 cash prize.
- **Eureka 3.0:** Best Innovation Award — ■2,000 cash prize.
- **TechXplore 25:** 1st Place — ■2,000 cash prize.
- **National IMPACTX Innovation Hackathon:** Top 3 — ■4,000 cash prize.
- **Yi Yuva Future 5.0:** Runner-Up.
- **Department Top Performer:** Artificial Intelligence & Data Science.

ADDITIONAL STRENGTHS

Technical communication • Workshop delivery • Curriculum planning • Practical demonstrations • Student engagement • Project reviews • Collaborative problem-solving • Technical documentation